SWARUP KUMAR SUBUDHI

Ph.D., Mechanical Engineering · University of Maryland, College Park
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RESEARCH INTERESTS

Printed and flexible electronics · Additive manufacturing for electronics and structural applications · Soft magnetic nanocomposites · Reliability engineering and physics of failure · Interfacial and magnetohydrodynamic fluid mechanics · Wearable and lab-on-chip sensing technologies · Advanced materials characterization · Process qualification for space and defense hardware

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EDUCATION

Ph.D., Mechanical Engineering 2026
University of Maryland, College Park, MD, USA · GPA: 3.7 / 4.0
Dissertation: Magnetic Films & Drops, and Their Interactions with Soft Materials
Advisor: Prof. Siddhartha Das, Department of Mechanical Engineering

Key Courses: Probabilistic Physics of Failure & Accelerated Testing, Failure Mechanisms of Electronic Packages, Composite Materials, Modelling Material Behavior, Mechanical Fundamentals of Electronic Components, Interfacial Fluid Mechanics

M.S., Mechanical Engineering 2024
University of Maryland, College Park, MD, USA
Graduate Teaching Assistant: Fluid Mechanics (ENME331), Engineering Mechanics I (ENES102)

B.Tech., Mechanical Engineering 2016
National Institute of Technology (NIT), Rourkela, Odisha, India
Thesis: Synthesis and Analysis of a Reconfigurable Manipulator Assembly Based on Five-Bar Linkage

Key Courses: Kinematics & Dynamics of Machinery, Machine Design I & II, Tool Design, Mechatronics, Digital Logic Design, Materials Technology, Mathematics (Minor)

RESEARCH EXPERIENCE

Graduate Research Assistant Jan 2022 – Present
Department of Mechanical Engineering, University of Maryland, College Park, MD

Doctoral Dissertation — Magnetic Films & Drops, and Their Interactions with Soft Materials

PI & Sponsor: Dr Siddhartha Das, Jan 2022 – Dec 2026

- Experimentally demonstrated magnetohydrodynamic de-encapsulation of superparamagnetic nanoparticle (MNP) aggregates from the interior of a sessile droplet using external magnetic fields; characterized aggregate morphology, size distribution, and extraction kinetics
- Demonstrated selective cargo transfer of MNP aggregates and encapsulated payload between adjacent droplets without physical contact — a key enabler for open-microfluidic lab-on-chip drug delivery platforms
- Fabricated free-standing PDMS–Fe_3O_4 soft magnetic nanocomposite films: synthesized and characterised oleic-acid-coated Fe_3O_4 MNPs (TEM, zeta-potential, VSM), dispersed in PDMS matrix, and characterized magnetic properties (VSM, hysteresis), mechanical flexibility (DMA), and RF/EMF shielding performance
- Investigated wetting dynamics and contact-angle evolution of MNP-laden drops on hydrophobic/hydrophilic surfaces as a function of MNP concentration, magnetic Bond number, and external field orientation
- Developed Python (OpenCV/CV2) image processing algorithms for automated droplet tracking, aggregate volume estimation from high-frame-rate videos, and point-cloud reconstruction of MNP aggregate 3D morphology
- Studied drying dynamics of particle-laden drops: ring-stain formation, Marangoni flows, and 3D deposit morphology as functions of particle concentration and substrate wettability (in collaboration, published Langmuir 2023)

Project 1: Development of Additively Manufactured High-Energy X-Ray Photoelectric Polarimeter

Sponsor: NASA-Goddard Space Flight Center · Jan 2024 – Dec 2026
PI: Dr. Siddhartha Das (UMD), Dr. Margaret Samuels (GSFC) & Dr. Abhijit Dasgupta (UMD)

- Achieved ±10 µm print tolerances for high-density (50–100 µm pitch) silver-nanoflake interconnects on space-grade substrates by direct-write extrusion printing, enabling a compact X-ray polarimeter assembly qualified for launch environments
- Developed novel extrusion toolpaths for printing on irregular and curved substrate geometries; systematically tuned ink rheology (viscosity, surface tension, particle size distribution) to eliminate satellite drops and bridging defects
- Designed and executed vibration qualification test matrices (launch-load profiles per NASA GEVS) for printed specimens; established pass/fail criteria and authored qualification documentation for program reviews
- Conducted environmental reliability tests (thermal exposure, humidity) on printed interconnects; characterized resistance drift and adhesion degradation as functions of thermal history
- Performed dimensional metrology (optical microscopy, SEM, surface profilometry) to verify print fidelity and cross-section quality of multi-layer silver and dielectric stacks

Project 2: Thermal Cycling Reliability of Aerosol-Jet and Extrusion-Printed Electronics

Sponsor: Laboratory of Physical Sciences, June 2023 – Dec 2023

- Built parametric multi-layer finite element models (ANSYS Mechanical, Abaqus CAE) of printed dielectric (NEA121, BCB) and conductive (sintered silver) stacks under thermal cycling (−40 °C to +140 °C, JEDEC JESD22-A104)
- Identified failure initiation sites via inelastic strain accumulation (Anand viscoplasticity model) and cyclic creep in sintered silver; predicted crack propagation paths in agreement with physical cross-sections
- Validated FEA predictions against experimental failure analysis: SEM imaging, EDS elemental mapping, and optical cross-section microscopy of cycled specimens
- Characterized material parameters (CTE, elastic modulus, creep coefficients) using TMA, DMA, and DSC for accurate model inputs

Project 3: Reliability Validation of Direct-Write Printed RF Devices

Sponsor: NextFlex in collab with Binghampton University & Lockheed Martin, Jan 2022 – May 2023

- Fabricated Metal-Insulator-Semiconductor (MIS) capacitors and coplanar waveguides (CPW) by extrusion direct-write printing of dielectric (NEA121) and silver paste on flat and curved substrates
- Developed custom print-head trajectory algorithms to achieve conformal deposition on non-planar surfaces; demonstrated functional RF prints on cylindrical and doubly-curved geometries
- Characterized RF performance using vector network analyzer (VNA): S-parameters (S11, S21), insertion loss, reflection coefficients, and quality factor Q across 0.1–6 GHz
- Qualified specimens against program specs for RF performance retention after thermal and mechanical stress; documented failure modes and process improvement recommendations

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Graduate Teaching Assistant Jan 2021 – Dec 2021
Department of Mechanical Engineering, University of Maryland, College Park, MD

- Instructed lab and studio sessions Fluid Mechanics (ENME331) and Engineering Mechanics I (ENES102)
- Held weekly office hours; graded assignments, quizzes, and exams for classes of 80+ students
- Developed supplementary worked-example problem sets for fluid statics, pipe flow, and rigid body dynamics

PROFESSIONAL EXPERIENCE

Senior Manager, Manufacturing — Transmissions 2019 – 2020

Tata Motors Ltd., CVBU, Jamshedpur, Jharkhand, India

Six Sigma: Warranty Cost Reduction — Gearbox Bearing Failures

- Led a cross-functional Six Sigma project (NPI, QA, SCM, R&D, Warranty) targeting bearing failures across the G600/750 and G1150 gearbox families in commercial vehicle transmissions
- Achieved ~50% reduction in IPTV/EPV for bearing failures (24-MIS warranty window) through systematic root-cause analysis, supplier process audits, incoming inspection protocol upgrades, and assembly control improvements
- Realized annual warranty savings of ₹4.53 Cr for the LKP2518 family of heavy commercial tipper trucks
- Applied DoE, Gauge R&R, SPC (Cpk), ANOVA, and Ishikawa tools throughout the DMAIC cycle; authored full project documentation and presented to senior management

Operational Line Efficiency Improvement — G950 Gearbox Assembly Line

- Improved Overall Line Efficiency (OLE) of the G950 gearbox assembly line by ~12% through systematic takt-time re-balancing and elimination/merging of low-performing bottleneck stations
- Deployed PFMEA-driven quality control checkpoints at defect-generating stations, achieving a measurable reduction in in-process defects
- Conducted RULA (Rapid Upper Limb Assessment) ergonomic assessments at operator-intensive stations; redesigned tooling and fixture heights to reduce operator fatigue and associated defect rates
- Implemented a low-cost IoT-based ANDON monitoring system integrating with existing line PLCs to capture real-time assembly losses; deployed Ishikawa and 5-Why corrective action plans for identified loss causes

Value Engineering

- Case study on hydraulic tipping kit of heavy commercial tippers: alternate sourcing and design modifications for gear pumps; documented cost-saving roadmap
- Case study on G1150 (9-speed gearbox) synchronizing parts: application of new material/surface treatment to improve cone performance and durability; reduced warranty cost for synchronizing-part failures

Assistant Manager, New Product Introduction (NPI) — Transmissions 2016 – 2019

Tata Motors Ltd., CVBU, Jamshedpur, Jharkhand, India

TATA Kestrel 8×8— Auxiliary Gearbox Launch

- Led NPI for the auxiliary transmission (transfer case / auxiliary gearbox) for the TATA Kestrel 8×8, a 4th-generation amphibious infantry combat vehicle
- Managed full product lifecycle from design review through proto build, supplier development, DFA/DFM analysis, incoming quality inspection, assembly, integration testing, and standardization; achieved 100% delivery adherence at each proto stage
- Developed and maintained testing matrix for gearbox performance, NVH tests, and durability; coordinated with R&D, Supplier Quality, and Army acceptance teams for technical sign-off
- Integrated auxiliary gearbox into Product Lifecycle Management (PLM) system; managed engineering change management throughout development

New Gearbox Launches — Commercial & Defense

- Successfully launched 10 new gearbox variants (including 6x6 and 8x8 defense truck transmissions) from development stage into mass production
- Managed supplier development and qualification for new auxiliary gearbox components; conducted first-article inspections, dimensional audits, and process capability (Cpk) assessments
- Maintained 100% on-time production adherence throughout the NPI period despite parallel launches across multiple gearbox families

PUBLICATIONS

Journal Articles — First Author

[1] **Subudhi, S.K.**, Zhao, B., Wang, X., Ting, J., Takeuchi, I., Dasgupta, A., and Das, S. (2024). Flexible and twistable free-standing PDMS–magnetic-nanoparticle-based soft magnetic films with robust magnetic properties. *Flex. Print. Electron.*, 9(1), 015013. https://doi.org/10.1088/2058-8585/ad2480

[2] **Subudhi, S.K.**, Chandel, G.R., Sivasankar, V., and Das, S. (2024). Magnetic nanoparticle aggregation and complete de-encapsulation of such aggregates from a liquid drop interior. *ACS Appl. Mater. Interfaces*, 16(46), 64253–64263. https://doi.org/10.1021/acsami.4c14135

[3] **Subudhi, S.K.** and Das, S. (2023). Reliability of lab-on-a-chip technologies for wearable electronics: a perspective. Frontiers in Sensors, 4, 1283402. https://doi.org/10.3389/fsens.2023.1283402

[4] **Subudhi, S.K.**, Madhira, S.A., and Sas, S. (2026). Drop-to-Drop Transfer of Magnetic Activity: Imparting Significant Heterogeneity and Reactivity to a Drop via Functionality Transfer from Another Drop. *ACS Appl. Mater. Interfaces*. https://doi.org/10.1021/acsami.6c13139

Journal Articles — Contributing Author

[5] Zhao, B., Sivasankar, V., **Subudhi, S.K.**, Sinha, S., Nikfarjam, S., Woehl, T., and Das, S. (2023). Three-dimensional deposits from drying particle-laden drops. Langmuir, 39(28), 9773–9784. https://doi.org/10.1021/acs.langmuir.3c00768

[6] Zhao, B., Sivasankar, V.S., **Subudhi, S.K.**, Dasgupta, S., and Das, S. (2023). Printed carbon nanotube-based humidity sensors deployable on surfaces of widely varying curvatures. ACS Applied Nano Materials, 6(2), 1459–1474. https://doi.org/10.1021/acsanm.2c04873

[7] Zhao, B., Sivasankar, V.S., **Subudhi, S.K.**, Sinha, S., Dasgupta, A., and Das, S. (2022). Applications, fluid mechanics, and colloidal science of carbon-nanotube-based 3D-printable inks. Nanoscale, 14(40), 14858–14894. https://doi.org/10.1039/D2NR04151F

[8] Dhamsania, H.J., et al. [**incl. Subudhi, S.K.**] (2022). Physically soft magnetic films and devices: fabrication, properties, printability, and applications. Journal of Materials Chemistry C, 10(17), 6563–6589. https://doi.org/10.1039/D1TC05500G

[9] Zhao, B. et al. [**incl. Subudhi, S.K.**] (2026). Fabrication, Characterization, and Reliability Analysis of 3D Printed High Frequency, Wideband, and Flexible Carbon-Nanotube and Graphene-Oxide Based Rectangular Patch Antennas Deployable on Curvilinear and Deformable Surfaces. Accepted for publishing in *Frontiers in Antennas and Propagation.*

Conferences – Contributing Author

[10] Janamanchi, P., **Subudhi, S.**, Dasgupta, A., and Quinn, E. (2025). Thermal Cyclic Fatigue Reliability of Printed Hybrid Electronics (PHE). *2025 26th International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems (EuroSimE)*, Utrecht, Netherlands, 2025, pp. 1-4, doi: 10.1109/EuroSimE65125.2025.11006613.

[11] Zhao, B. *et al.*, "3D Printed Wideband Flexible Carbon-Nanotube and Graphene-Oxide Based Rectangular Patch Antennas Deployable on Deformable Surfaces," *2026 IEEE International Conference on Flexible and Printable Sensors and Systems (FLEPS)*, Atlanta, GA, USA, 2026, pp. 1-4, doi: 10.1109/FLEPS69128.2026.11645063

TECHNICAL SKILLS

Additive Manufacturing & Fabrication

Direct-Write Printing: Extrusion printing (nScryptPro, Hyrel3D), Inks: Silver pastes, dielectric inks, CNT-GO ink formulation and characterization, Polymers (PDMS, PLA)

3D Printing: FDM (Ultimaker, Hyrel, CraftPro, Prusa & Formlabs platforms), SLA/DLP resin printing; print parameter optimization for dimensional accuracy

Subtractive & Traditional: CNC milling/turning, laser cutting/tooling, mold development, injection molding concepts

CAM Software: Autodesk Fusion 360, SolidWorks CAM

Computer-Aided Design & Simulation

CAD: SolidWorks (advanced — assemblies, drawings, simulation), Creo Parametric (Pro-E), AutoCAD

FEA / CAE: ANSYS Mechanical, ANSYS Workbench, ANSYS Icepak (thermal), Abaqus CAE (non-linear, visco-plasticity, creep)
CFD & Multiphysics: ANSYS Fluent, COMSOL Multiphysics (fluid-structure, electromagnetic, heat transfer), python-based Multiphysics Particle Simulator

Reliability Engineering & Testing

Accelerated Testing: Thermal cycling (JEDEC JESD22-A104), temperature-humidity bias (THB, JESD22-A101), vibration/shock testing (GEVS STD-7000B), deformational loading (most standards)
Failure Analysis: SEM/EDS fractography, optical cross-section microscopy, resistance monitoring, Physics of Failure (PoF) modelling, Weibull analysis, acceleration factor estimation
Quality & Process: Six Sigma (DMAIC), DoE (full factorial, RSM), PFMEA, DFA/DFM, ANOVA, SPC (Cpk), MSA / Gauge R&R, NPI process, PLM, BOM management

Materials Characterization

Thermal: TMA (CTE measurement), DSC (Tg, melting, crystallization), TGA (decomposition), DMA (storage modulus, loss factor, creep)
Microscopy & Surface: SEM (morphology, cross-section), EDS (elemental mapping), optical microscopy, laser scanning confocal microscopy, surface profilometry (stylus, white-light interferometry)
Rheology (inks & fluids): Viscosity (cone-plate, parallel-plate rheometry), surface tension (pendant drop), zeta-potential (DLS), particle size distribution (DLS, SEM-based)
Magnetic: VSM (magnetization curves, saturation magnetization, coercivity), hysteresis loop characterization
Dimensional: GD&T, CMM concepts, optical comparator, digital calipers/micrometers, MSA

Computation & Data Analysis

Scientific computing: MATLAB & Simulink (optimization toolbox, image processing toolbox, GUI development), Wolfram Mathematica, R / RStudio
Programming: Python (image processing, point tracking, data analysis), C, C++, Arduino IDE
Web & Scripting: HTML5, CSS3, JavaScript
Data & Reporting: JupyterLab, Microsoft Office (Excel, PowerPoint, Word), Power BI, Origin, Minitab

Instrumentation & Control

Control systems: Arduino (sensor integration, actuator control), MATLAB-based closed-loop control, stepper/servo motor control
Electronics: Circuit design (through-hole and SMD), PCB layout concepts, oscilloscope/multimeter/LCR meter, vector network analyser (VNA) operation
Sensors: Strain gauges, thermocouples, load cells, optical encoders, Hall-effect sensors, MEMS accelerometers

ACADEMIC PROJECTS

Synthesis & Applications of Reconfigurable Manipulator Assembly based on a 5-Bar Linkage

B.Tech. Thesis, 2016, NIT Rourkela
Advisor: Prof. J. Srinivas

- Synthesized and fabricated a 5-bar linkage reconfigurable manipulator with switchable end-effector trajectories
- Performed kinematic and dynamic analysis (forward/inverse kinematics, Jacobian, workspace mapping) in MATLAB
- Integrated closed-loop motor control via Arduino; developed a MATLAB GUI for user-friendly trajectory input, real-time visualization, and path optimization

Optimization of Taper Angle of Structures Machined by EDM Processes

Undergraduate Project, 2015–16, NIT Rourkela
Advisor: Prof. S. Gangopadhyay

- Applied Response Surface Methodology (Box-Behnken design) to optimize Material Removal Rate (MRR) and taper angle in reverse electro-discharge machining (R-EDM)
- Performed ANOVA to quantify the effects and interactions of current, pulse-on time, and electrode rotation speed; developed regression model with $R^2 > 0.95$

Combustion Modelling — Diesel Engine on Carbon Black Slurry
Undergraduate Project, 2015, NIT Rourkela
Advisor: Prof. S. Murugan

- Developed a two-zone thermodynamic combustion model for a DI diesel engine running on carbon black–water–diesel slurry fuel
- Predicted in-cylinder pressure, temperature, heat release rate, and emissions as functions of blend ratio; validated against experimental cylinder pressure traces

Electric Karting Vehicle — Chassis & Structural Design Team Project, 2014–15

- Designed and fabricated chassis for a single-seat electric kart; performed FEA (Abaqus) for structural deformation and stress under design loads
- Served as mechanical designer for drive train integration; validated working model against design constraints

PRESENTATIONS & PROFESSIONAL SERVICE

Conference & Symposium Presentations

- Subudhi, S.K. Capillary Force Driven Extraction of MNPs in High-Viscous Mediums. Presented at GWU Symposium, Washington D.C. (2025)
- Subudhi, S.K. et al. Soft magnetic PDMS–nanoparticle films: fabrication and characterization. Presented at UMD GRAD, College Park, MD (2023)
- Cost-Optimization Case Study on a Hydraulic Tipping Kit. 35th INVEST Conference, Kolkata, IN, Nov 2019.

Professional & Academic Leadership Roles

- ENME Representative, Graduate Student Government (GSG), University of Maryland — Jul 2022 – May 2025
  - Elected departmental representative; served as Chair of Academic Affairs Committee, advocating for graduate student academic policy at the university level
- Executive Body Member, Mechanical & Reliability Engineering Graduate Association (MEGA), University of Maryland — Jul 2024 – Present
- Cross-functional team leader (3–5 engineers), Tata Motors Ltd. — 2018–2020
- Assembly Team Leader (2-3 floor managers, 30-40 operators), Tata Motors ltd. — 2019–2020

WORK AUTHORIZATION

F-1 student visa holder (India). Eligible for 12-month OPT + 24-month STEM OPT extension upon graduation, providing up to 36 months of US work authorization. Employer requires E-Verify enrollment only — no H-1B petition, lottery, or attorney fees required for the initial employment period. Available to start mid-2026.

REFERENCES

Available upon request. Primary academic reference: Prof. Siddhartha Das, Department of Mechanical Engineering, University of Maryland, College Park (sidd@umd.edu).